Mohamed Emam

Biomechanics Researcher · AHPRA-Registered Physiotherapist

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PROFESSIONAL SUMMARY

Biomechanics researcher and AHPRA-registered physiotherapist completing a joint PhD at Semmelweis University (Budapest) and Cairo University (defence: November 2026). I bridge clinical rehabilitation with cutting-edge biomechanics using markerless motion capture (OpenCap) and statistical parametric mapping. Nine years of clinical and academic experience, 8 peer-reviewed publications in Q1/Q2 journals, and full general registration with AHPRA. Research spans gait analysis, neurorehabilitation, electrical stimulation, and aging physiology. Seeking postdoctoral fellowship opportunities.

EDUCATION

Joint PhD in Health Sciences — 2022 – Present
Semmelweis University, Budapest & Cairo University, Egypt
Markerless motion capture, gait analysis & rehabilitation biomechanics. Defence expected November 2026.

MSc in Physical Therapy — 2015 – 2019
Cairo University, Egypt

BSc in Physical Therapy (Honours) — 2008 – 2013
Cairo University, Egypt

PROFESSIONAL EXPERIENCE

Assistant Lecturer — 2016 – Present
Kafr El-Sheikh University, Egypt
Teaching musculoskeletal physiotherapy, lab demonstrations, curriculum development, student research mentorship.

Founder & Physiotherapist — 2015 – Present
Start-Rehabilitation Centre, Kafr El-Sheikh
Clinical management of musculoskeletal & orthopaedic conditions; manual therapy and exercise rehabilitation; clinic operations.

R&D Team Lead — 2022 – 2025
Physioway (Physiotherapy Technology Company)
Directed development of clinical applications (scoliosis assessment tool, vestibular rehabilitation app).

SELECTED PUBLICATIONS

1. **Impact of Russian Current Therapy Targeting the Anterior Tibial Group on Balance and Fall Risk in Post-Stroke Patients: An RCT.** *Neurological Sciences*, 2026.
2. **Wrist Muscle Performance in Students with Chronic Nonspecific Neck Pain: An Isokinetic Assessment.** *BMC Musculoskeletal Disorders*, 2026.
3. **Effect of Gaze Direction Recognition Task on Pain, ROM and Functional Activities in Cervicogenic Headache Patients.** *BMC Neurology*, 2026.
4. **Proprioceptive Training Reduces Headache Burden and Centre of Pressure Path Length in Cervicogenic Headache: An RCT.** *Physiology International*, 2025.

5. **Effect of Different Types of Electrical Stimulation on Postural Control and Gait After Stroke: A Systematic Review and Meta-Analysis.** *Physiology International*, 112(4), 2025. Q2
6. **Blood Flow Restriction Exercise Protocols for Elderly Populations: A Scoping Review.** *Journal of Clinical Medicine*, 14(12):4185, 2025. Q1
7. **Proprioceptive Training Improves Postural Stability and Reduces Pain in Cervicogenic Headache Patients: An RCT.** *Journal of Clinical Medicine*, 13(22):6777, 2024. Q1
8. **Effect of Cervicogenic Headache on Myoelectrical Activities of Suboccipital Muscles, ROM and Functional Activities at Different Ages.** *Medical Journal of Cairo University*, 2019.

TECHNICAL SKILLS

- **Motion capture:** OpenCap (markerless), marker-based systems
- **Statistics:** SPM1D, Linear Mixed-Effects Models
- **Programming:** MATLAB, Python, R, SPSS
- **Imaging:** MRI muscle segmentation & analysis
- **Assessment:** Isokinetic dynamometry, surface EMG
- **Design:** RCTs, systematic reviews, meta-analyses

AWARDS, REGISTRATION & LANGUAGES

- **AHPRA** — Physiotherapist, Full General Registration (No. PHY0002905130)
- **Scholarships:** Stipendium Hungaricum · Global Korea Scholarship · SE 250+ Excellence PhD · EKOP Doctoral Excellence Grant
- **Recognition:** Poster Award, 1st International Health Science Conference, Szeged (2025); 5 peer reviews completed
- **Languages:** Arabic (native) · English (C1–C2, PTE 81) · Hungarian (A1)